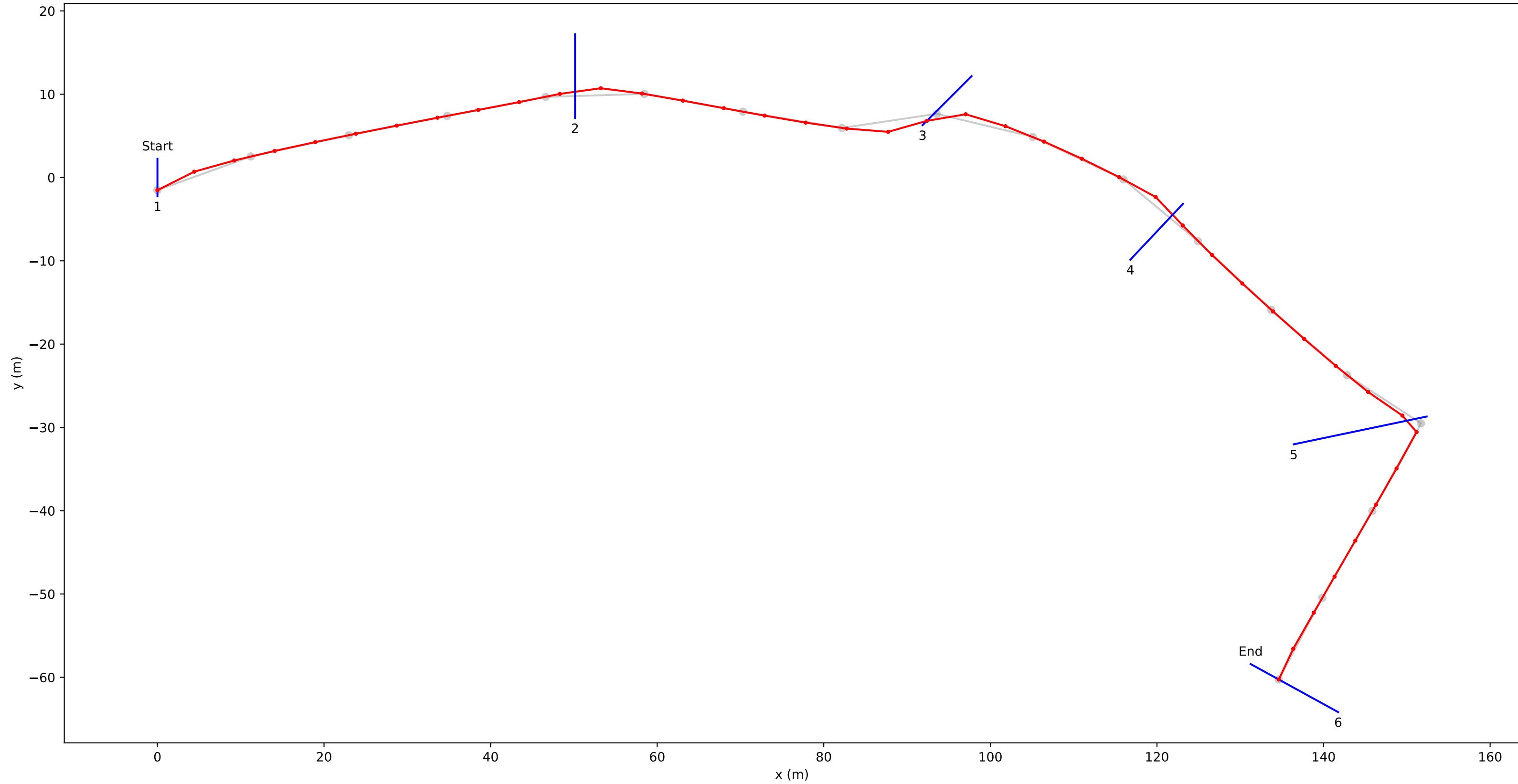


The figure displays a 2D plot of a trajectory. The horizontal axis is labeled 'x (m)' and ranges from 0 to 160. The vertical axis is unlabeled. A red line represents the trajectory, with points 1 through 6 marked. Point 1 is labeled 'Start' and point 6 is labeled 'End'. A grey line represents a reference path. Blue lines are drawn tangent to the red trajectory at points 1, 2, 3, 4, 5, and 6. The trajectory starts at point 1, moves upwards and to the right, then curves downwards and to the right, ending at point 6. The reference path is a straight line from point 1 to point 6. The blue tangent lines show the direction of the trajectory at each point.



Track Generation Report

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Generated: 2026-05-30 14:53:16

Gate file: ref_gates_gates.csv

Gate count: 6

Optimization parameters

method: L-BFGS-B

maxiter: 1000

ftol: 1e-07

Point counts

objective function points: 42

path generation points: 18

Results

total optimization time: 14.238 s

total path length: 204.5700 m

Per-step time estimate (one measured objective evaluation)

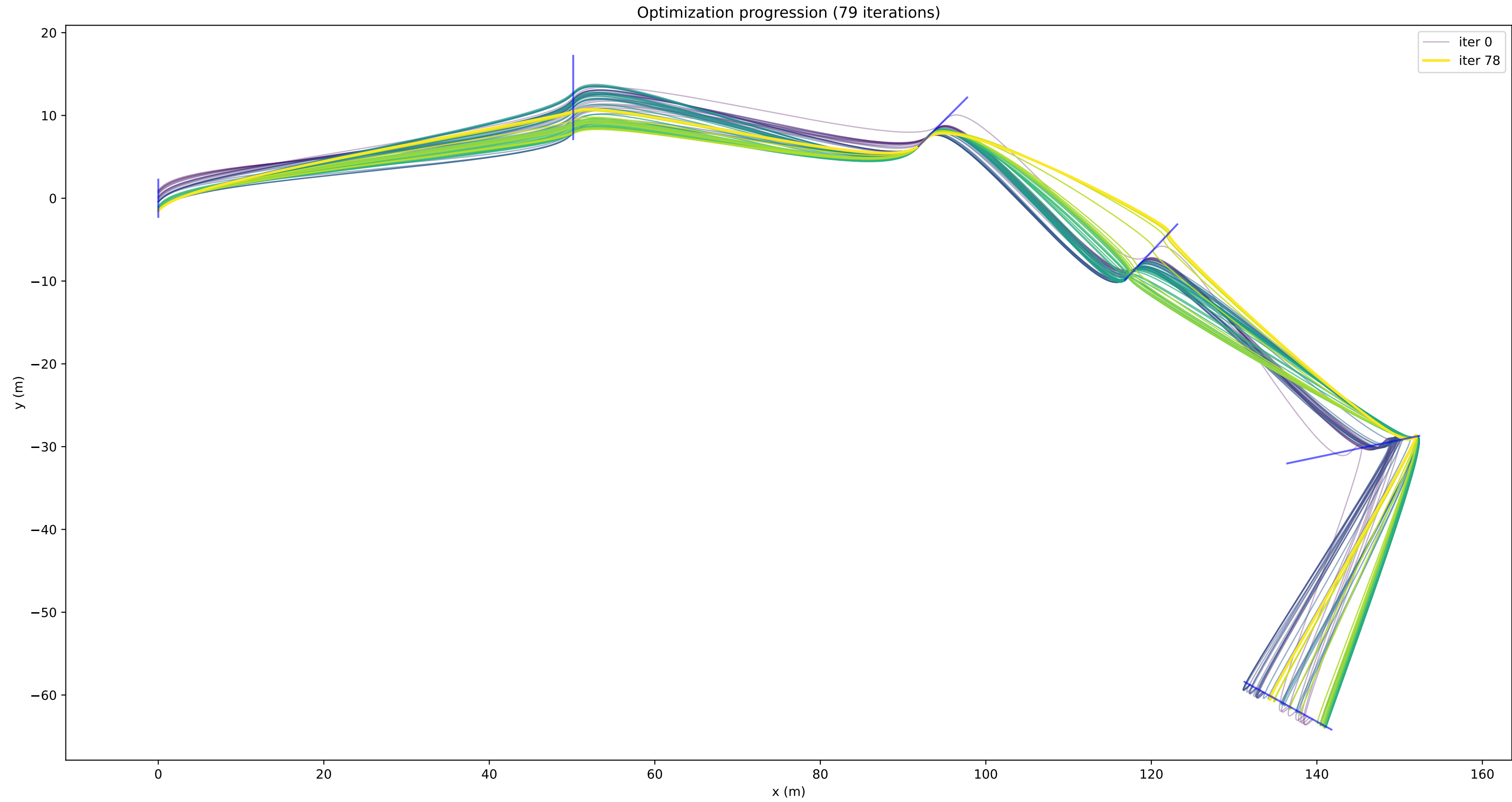
make spline from dofs: 0.98 ms

path length function: 0.49 ms

equidistant times generated: 0.33 ms

curvature objective function: 1.69 ms

Optimization progression



Equally spaced iterations

5 equally spaced iterations

